

SMALL CLUSTERS OF THE HAT AND SPECTRE DO NOT TILE THE PLANE

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ABSTRACT. Smith, Myers, Kaplan and Goodman-Strauss discovered the Hat family of aperiodic monotiles and the strictly chiral Spectre family. We ask whether a connected union of several congruent tiles from either family can itself tile the plane monohedrally, with reflected cluster copies allowed. We give a parameter-transfer argument for the straight-edged Hat family and exact finite calculations through five tiles in both families. No such cluster admits two complete coronas. Consequently, no topological-disc cluster of two through five Hat-family tiles or Spectres tiles the plane periodically or aperiodically. The calculations allow arbitrary proportions of direct and reflected cluster copies. They use integer kite-grid arithmetic for the Hat family and arithmetic in $\mathbb{Q}(\sqrt{3})$ for Spectres. This draft records a computer-assisted result pending an independent implementation and compact rejection certificates.

1. INTRODUCTION

An aperiodic monotile is a planar shape which tiles the plane, while every tiling by its congruent copies is non-periodic. Smith, Myers, Kaplan and Goodman-Strauss found a continuum of straight-edged examples containing the Hat and Turtle [1]. They later found the Spectres, a family of strictly chiral aperiodic monotiles [2].

Each published proof groups tiles into larger patches or metatiles. This suggests a different question: can a fixed finite cluster of Hats or Spectres tile the plane using congruent copies of the cluster? If it can, splitting each cluster copy gives a tiling by the original aperiodic tile. A periodic cluster tiling would admit a periodic choice of these dissections. Thus any tileable cluster would be another aperiodic monotile in the same lineage.

This paper settles the first four non-trivial cluster sizes in both families. Its main results are computer-assisted.

Theorem 1.1 (Hat-family clusters). *Let $a, b > 0$ with $a \neq b$. No topological-disc union of two through five congruent copies of $\text{Tile}(a, b)$ tiles the plane monohedrally, even when reflected cluster copies are allowed.*

Theorem 1.2 (Spectre clusters). *Let S be a Spectre for which every tiling is one-handed and straightens equivariantly to a vertex-to-vertex tiling by $\text{Tile}(1, 1)$. No topological-disc union of two through five congruent copies of S tiles the plane monohedrally, even when reflected cluster copies are allowed.*

The hypotheses of Theorem 1.2 include the smooth and C^1 Spectres of [2]. They also include any polygonal construction for which the same alignment and straightening theorem has been proved.

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The calculations actually cover all edge-connected clusters in the Spectre case, including some unions which are not discs. A union joined only at isolated points has disconnected interior and lies outside the conventional topological-disc monotile setting.

2. CLUSTERS AND CORONAS

Definition 2.1. A cluster of size n is the union of n congruent base tiles with disjoint interiors. It is edge-connected if the adjacency graph formed by shared boundary arcs is connected. Two clusters are equivalent when a Euclidean isometry maps one union to the other.

Throughout this paper, “congruent copies” permits every Euclidean isometry, including reflections. Thus a tiling may use a cluster and its mirror image in arbitrary proportions. A result under this convention also rules out tilings which use direct isometries only.

A complete first corona around a central cluster is a finite collection of non-overlapping congruent clusters which covers its entire boundary. Having $k + 1$ coronas means that the exposed boundary after k layers also has a complete cover.

Lemma 2.2 (Corona obstruction). *If a compact cluster tiles the plane, then it admits complete coronas of every finite depth.*

Proof. Choose one copy in a plane tiling. Local finiteness gives the finite set of copies meeting its boundary. Repeat with the exposed boundary of the finite patch. This constructs every prescribed finite number of layers. $\square$

It follows that failure at any finite corona depth proves that a cluster does not tile the plane.

3. TRANSFER ACROSS THE HAT FAMILY

Write $T_r = \text{Tile}(1, r)$. The endpoints $r = 0, \infty$ and the equilateral value $r = 1$ admit periodic tilings. We consider the aperiodic parameters $0 < r < \infty$, $r \neq 1$.

Smith et al.’s parameter theorem gives a bijection between the tilings by T_r and T_s for any two aperiodic parameters [1, Theorem 6.1]. It preserves tile orientations and adjacencies while changing one edge length.

Proposition 3.1 (Cluster transfer). *Fix a finite patch type. Its realisation by T_r tiles the plane by congruent cluster copies if and only if its realisation by T_s does.*

Proof. Split a cluster tiling into its constituent T_r tiles and apply the tiling bijection. The deformation commutes with every isometry used to place a cluster copy. Hence all copies become congruent copies of one T_s cluster. The inverse deformation proves the converse. $\square$

It therefore suffices to calculate at the Hat parameter $r = \sqrt{3}$. Every Hat in a tiling aligns with one [3.4.6.4] kite grid. A Hat is a union of eight kite cells, so all contact, overlap and corona tests use integer coordinates.

4. REDUCTION FOR SPECTRES

Split a hypothetical tiling by Spectre clusters into individual Spectres. The chirality theorem says that all individual tiles have one hand, even when reflections were initially allowed [2]. Fixing that hand reduces every cluster placement to a direct isometry.

Straightening every curved edge produces a vertex-to-vertex tiling by $\text{Tile}(1, 1)$ and commutes with the cluster isometries. The fourteen straight edges have unit length. An even curved edge can meet only an odd curved edge. All possible contacts therefore arise by aligning an even and an odd edge of two straight tiles and rejecting interior overlap.

The vertices of $\text{Tile}(1,1)$ lie in $\mathbb{Q}(\sqrt{3})^2$. The calculation uses exact pairs $a + b\sqrt{3}$, with rational a, b . Signs are decided by comparing a^2 with $3b^2$, so no floating-point tolerance enters a geometric decision.

5. A FREQUENCY OBSTRUCTION

Both tiling spaces have two distinguished tile classes. For the Hat family they are the two hands, whose frequency ratio is $\varphi^4 : 1$, where $\varphi = (1 + \sqrt{5})/2$. For Spectres they are orientations at even and odd multiples of 30 degrees, whose ratio is $(4 + \sqrt{15}) : 1$. Reflection swaps the Hat classes, while an odd multiple of 30 degrees swaps the Spectre classes.

Lemma 5.1 (Two-class frequency bound). *Suppose a cluster of size n contains p base tiles in one class and $q = n - p$ in the other. If copies of the cluster tile the plane, and the smaller base-tile frequency is f , then*

$$\min(p, q) \leq nf.$$

Proof. Let x be the frequency of cluster copies placed by isometries which preserve the two classes. The frequency of the first base-tile class is $(xp + (1 - x)q)/n$. It lies between p/n and q/n . The complementary frequency does too, with the endpoints exchanged. The smaller prescribed frequency is therefore at least $\min(p, q)/n$. $\square$

For Hats, $f = 1/(\varphi^4 + 1) < 1/7$. Hence every candidate cluster of size at most seven consists entirely of one hand, up to reflecting the whole cluster. For Spectres, $f = 1/(5 + \sqrt{15}) < 1/8$. Every candidate cluster of size at most eight therefore uses only one orientation parity, up to rotating the whole cluster by 30 degrees. This lemma does not settle those sizes, but it gives a strong exact filter for the next computations.

6. FINITE ENUMERATION

For each family, first enumerate every directed legal contact of two base tiles. Build a connected cluster by adjoining one legal neighbour at a time. Canonicalise after every addition under the relevant Euclidean symmetry group.

Every connected cluster has a spanning tree. Canonical augmentation adjoins its leaves one at a time, so the construction enumerates every edge-connected cluster through size four. At size five, the frequency lemma permits the Hat search to retain only one-handed clusters and the Spectre search to retain only one-parity clusters. The Hat calculation also retains contacts and clusters with cavities or cut points.

For every cluster, enumerate all congruent neighbouring copies by aligning constituent boundary contacts. An exact-cover backtracker chooses non-overlapping copies which cover the exposed boundary. It always branches on an uncovered cell or edge with the fewest available placements.

6.1. Hat-family result. There are 58 directed Hat contacts. Fifty-four have simply connected union and reproduce the contact count in the authors' validation model. Four more enclose a cavity and are retained here.

Size	Enumerated candidates	One corona	Two coronas
2	35	6	0
3	1096	6	0
4	46175	6	0
5	230411	1	0

The size-five row is the complete frequency-compatible census. Lemma 5.1 rules out every omitted pentamer. Of the eighteen one-corona survivors through size four, five satisfy the frequency condition.

Lemma 2.2 rules out all these clusters at the Hat parameter. Proposition 3.1 proves Theorem 1.1.

6.2. Spectre result. The exact straightened atlas contains 37 directed, non-overlapping even-to-odd contacts. Reflections are not used after the handedness has been fixed. We therefore report direct-congruence classes; retaining mirror pairs separately can only enlarge the finite search.

Size	Enumerated candidates	One corona	Frequency-compatible	Two coronas
2	19	4	2	0
3	438	2	1	0
4	11316	0	0	0
5	22042	0	0	0

The size-five row is the complete one-parity census. Lemma 5.1 rules out every omitted pentamer.

Lemma 2.2 proves Theorem 1.2.

In particular, neither theorem leaves a periodic case: the clusters do not tile the plane at all. They are therefore neither periodic monotiles nor aperiodic monotiles.

7. VERIFICATION STATUS

The source programs are `hat-family-clusters.py` and `spectre-clusters.py`. The independent Hat checker is `verify-hat-cluster-sat.py`. The complete summary calculations are:

```
python3 hat-family-clusters.py --dimers --coronas 2 --summary
python3 hat-family-clusters.py --trimers --coronas 2 --summary
python3 hat-family-clusters.py --tetramers --coronas 2 --summary --jobs 8
python3 hat-family-clusters.py --pentamers --coronas 2 --summary --jobs 8
python3 spectre-clusters.py --dimers --coronas 2 --summary
python3 spectre-clusters.py --trimers --coronas 2 --summary
python3 spectre-clusters.py --tetramers --coronas 1 --summary --jobs 8
python3 spectre-clusters.py --pentamers --coronas 2 --summary --jobs 8
```

Both programs use exact arithmetic and include internal calibration checks. The Hat program reproduces the published 54-contact simply connected atlas. The Spectre program checks all fourteen edge lengths and the 37-contact parity atlas.

A separate Hat verifier rebuilds the kite-grid geometry without importing the enumerator. It confirms that all nineteen recorded Hat survivors admit one corona, then encodes two coronas as SAT. CryptoMiniSat independently reports all nineteen formulas unsatisfiable. This checks the decisive local obstructions with a different search algorithm.

The result should remain labelled computer-assisted until a separate implementation reproduces the eight censuses and the Spectre refutations. The frequency bound eliminates sixteen of the twenty-five one-corona survivors. A publication bundle should also contain independently checkable SAT proof certificates for the nine frequency-compatible cases.

8. LARGER CLUSTERS

The next unresolved size is six in both families. The growth from 1096 three-Hat clusters to 46,175 four-Hat clusters shows that direct enumeration will soon become impractical. Spectre clusters grow from 438 to 11,316 classes.

Applying Lemma 5.1 during canonical augmentation leaves 230,411 one-handed Hat pentamers and 22,042 one-parity Spectre pentamers. Exactly one of the 230,411 Hat pentamers admits a first corona, and it admits no second corona. None of the 22,042 Spectre pentamers admits a first corona.

We stop the size-by-size enumeration at five. A direct extension to size six is possible but would give another bounded result at substantially greater cost. The next stage seeks a structural argument for arbitrary finite size.

An all-size proof would need a structural obstruction. Both original tiling spaces have recognisable hierarchies. A possible route is to track the states in which a finite cluster boundary crosses metatile boundaries. If every non-singleton state forces a non-singleton crossing at the next hierarchical level, a finite cluster would have to cross boundaries at arbitrarily large scales. That would be impossible. Recognisability alone does not give this claim, since cluster boundaries may cross supertile boundaries. Establishing a finite boundary-state statement is the main open problem.

Question 8.1. Does any finite edge-connected cluster of at least six Hats or Spectres tile the plane monohedrally?

REFERENCES

- [1] D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss, *An aperiodic monotile*, Combinatorial Theory **4** (2024), no. 1, <https://arxiv.org/abs/2303.10798>.
- [2] D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss, *A chiral aperiodic monotile*, Combinatorial Theory **4** (2024), no. 2, <https://arxiv.org/abs/2305.17743>.

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